

Mentor review package

HUMAN ACTION REQUIRED — ten items, G-1 through G-10.

Every one is a *confirmation* of something already computed. None asks for a calculation.

Decision form: `review/MENTOR_DECISION_FORM.md`.

Full item text: `spinor_trace_bridge/docs/MENTOR_REVIEW_ITEMS.md`.

Read in this order. The first three are the ones where a wrong answer changes a number rather than a sentence.

Priority 1 — G-10: the leading-degree rule

This is the only unverified analytic input the degree-ten result rests on.

The stress-flow closure assigns each generated target to a graded piece using a leading field degree per generator. The rule:

$$\begin{aligned}\mathrm{Tr}(\tau) &\longrightarrow 4, \text{ not } 2 \\ \mathrm{Tr}(\tau^k) &\longrightarrow 2k, \quad k \geq 2 \\ \text{products} &\longrightarrow \text{additive}\end{aligned}$$

All 18 generators at all six primes place their first appearance exactly where this predicts (`tests/test_leading_d`) so the assignments are no longer self-referential — they agree with a rule written down separately.

G-10 has since been derived, not merely checked against the table. For any improvement coefficient c the trace is $(1 - cd)\langle F, F \rangle$, so only $\langle F, F \rangle$ matters; $F \wedge F$ is a top form built from two copies of an odd-degree form and vanishes; $\langle F, \star F \rangle$ is proportional to it; hence $\langle F, F \rangle = 0$ on either eigenspace. Verified with a control at four primes by code importing no flow machinery, and load-bearing: forcing a degree-2 contribution gives $\dim Q_{10} = 0$ instead of 3.

What only you can confirm is now narrower: that the free theory and stress convention are the ones you intend, and that no formulation in use changes the *quadratic scalar available*. Independence from c means a different trace convention alone cannot change the conclusion.

If this is wrong: every target lands in the wrong graded piece, and $\dim D_{10} = 11$ — hence $\dim Q_{10} = 3$, the paper's central number — is wrong with it. Nothing else in the paper has this property.

Priority 2 — G-2: the split-signature correction

An earlier record in this project stated the oscillator frame's real form is Euclidean $SO(10)$, where $\star^2 = -1$ admits no real self-dual five-form, and treated that as blocking the bridge.

That was wrong. Computing the metric from the archive's own wedge and contraction operators — all one hundred anticommutators, diagonalised — gives eigenvalues $(+\frac{1}{2})^5, (-\frac{1}{2})^5$: real signature $(5, 5)$, split. A null frame cannot be Euclidean at all, since Euclidean signature has no isotropic vectors.

	real form	$\star^2$ on 5-forms	real self-dual 5-forms
Lorentzian	$\text{Spin}(1, 9)$	+1	yes
split	$\text{Spin}(5, 5)$	+1	yes
Euclidean	$\text{Spin}(10)$	−1	no
complexified	$\text{Spin}(10, \mathbb{C})$	—	the common ground

What survives is weaker and precise: $(5, 5)$ and $(1, 9)$ are inequivalent **real** forms, so a real frame transformation between them does not exist. But both metrics have discriminant -1 up to squares, so over $\mathbb{C}$ and over $\mathbb{F}_p$ they are congruent, and the bridge constructs that congruence explicitly and checks it. Every component-level comparison in the paper is therefore modular, where the transition exists exactly — never a real Lorentzian identification.

What is needed: confirmation that the correction is accepted, and — because the earlier, wrong statement may already have been communicated to you — that it has not been relied on elsewhere.

Priority 3 — G-8: credit for rank 81

The number 81 is **not ours** and the paper says so.

element	origin
analytic count $126 - 45 = 81$ from the trivial generic stabiliser	literature , cited
earlier float64 finite-difference Jacobian evidence	mentor archive
exact analytic amputated derivative, no step size, no tolerance	this project
integral basis making the modular rank a rigorous bound	this project
explicit 81×81 minor, two determinant routines	this project
complete 83/83 candidate schedule with terminal statuses	this project

The claim split in the draft: the *upper* bound is analytic and attributed; only the *lower* bound $\text{rank}_{\mathbb{Q}} \geq 81$ is claimed here, and only at finitely many sample points. A wording gate fails the build on “proved rank 81 computationally”.

What is needed: confirm the division of credit and that the literature attribution points at the source you intend. Also confirm you want the paper to state plainly that 83 candidates of rank 81 means the selection carries functional dependencies — it currently does.

Priority 4 — G-9: how to present the cardinality proposition

Any set closing degree d has at least $\dim Q_d$ elements, because the quotient map is linear and a span of k vectors has dimension at most k . Three lines.

It is in the paper because it discharges half of PO-08 and turns a basis-dependent assertion into a basis-free one — not because it is new. It is almost certainly not new as mathematics, and the draft makes no novelty claim.

Options: proposition with proof (current) · lemma · remark · corollary · omit as elementary.

Remaining items

item	subject	what a wrong answer costs
G-1	reconstructed spinor index placement	a wording change
G-3	real-form independence of dimensions	a citation instead of an argument
G-4	equivariance at sampled group elements	possibly a symbolic proof
G-5	comparison is modular, not char. zero	wording only
G-6	real-form caveat; AMB-01/02 source intent	wording; no result depends on it
G-7	redistribution of the spinor archive	release contents only

What is not being asked

No item asks you to approve a submission, an author list, or a novelty claim. Those are separate and are in `audit/AUTHORSHIP_AND_CREDIT_FINAL.md`. Every novelty row is `PROVISIONAL` and stays that way until someone who knows the field says otherwise — a literature search shows absence of evidence, not evidence of absence.